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Library-driven approach for fast implementation of the voxel spread function to correct magnetic field inhomogeneity artifacts for gradient-echo sequences

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Purpose: Previously developed Voxel Spread Function (VSF) method (Yablonskiy, et al, MRM, 2013;70:1283) provides solution to correct artifacts induced by macroscopic magnetic field inhomogeneity in the images obtained by multi-Gradient-Recalled-Echo (mGRE) techniques. The goal of this study was to develop a library-driven approach for fast VSF implementation.

Methods: The VSF approach describes the contribution of the magnetic field inhomogeneity effects on the mGRE signal decay in terms of the F-function calculated from mGRE phase and magnitude images. A pre-calculated library accounting for a variety of background field gradients caused by magnetic field inhomogeneity was used herein to speed up the calculation of F-function. Quantitative $R2^*$ maps from the mGRE data collected from two healthy volunteers were generated using the library as validation.

Results: As compared with direct calculation of the F-function based on a voxel-wise approach, the new library-driven method substantially reduces computational time from several hours to few minutes, while, at the same time, providing similar accuracy of $R2^*$ mapping.

Conclusion: The new procedure proposed in this study provides a fast post-processing algorithm that can be incorporated in the quantitative analysis of mGRE data to account for background field inhomogeneity artifacts, thus can facilitate the applications of mGRE-based quantitative techniques in clinical practices. © 2021 American Association of Physicists in Medicine [https://doi.org/10.1002/mp.14904]

Key words: fast calculation, gradient echo, library, $R2^*$, voxel spread function

1. INTRODUCTION

It is well-known that macroscopic B_0 magnetic field inhomogeneity adversely affect quantitative measurements of relaxation parameters obtained with multi-Gradient-Recalled-Echo (mGRE) MRI if not accounted for.

It was demonstrated recently that quantitative mGRE-based approach incorporating voxel spread function (VSF) method for field inhomogeneity correction¹ provides quantitative information on brain tissue cellular structure² as well as its alterations under healthy aging³ and different neurological

diseases.^{4–7} The VSF method takes advantage of utilizing both magnitude and phase information of mGRE data from a single scan, hence it saves acquisition time and avoids mis-registration errors. However, a direct implementation of the VSF method requires a large number of voxel-dependent convolutions resulting in long computational time.

In this study, we propose a library-driven approach for fast implementation of the VSF method. Instead of the direct voxel-wise calculation, a pre-calculated library accounting for a variety of background field gradients caused by magnetic field inhomogeneity is used. The library-driven

approach substantially reduces the computational time of VSF from hours to a few minutes, and provides similar accuracy compared with the direct calculation. This can facilitate the applications of mGRE-based quantitative techniques in clinical practices.

2. METHODS

2.A. Brief overview of the VSF method

As described previously,¹ MR signal in the presence of inhomogeneous magnetic field $b(\mathbf{r})$ can be represented as follows:

$$\tilde{S}(\mathbf{k}; TE) = \int d\mathbf{r} \cdot \rho(\mathbf{r}; TE) \cdot \exp[-2\pi i \mathbf{k} \mathbf{r} + i\gamma b(\mathbf{r})(TE) + i\varphi_0(\mathbf{r})] \quad (1)$$

where $\mathbf{k}$ and $\mathbf{r}$ represent the k space and image space, respectively, γ is the gyromagnetic ratio, $\varphi_0(\mathbf{r})$ is the signal phase shift at $TE = 0$. Here, we only considered the gradient echo time TE and ignore the time during gradient echo acquisition. By using a linear approximation, $b(\mathbf{r})$ and $\varphi_0(\mathbf{r})$ can be represented in the n th voxel as follows:

$$b_n(\mathbf{r}) = b_n + \mathbf{g}_n \cdot \mathbf{r} = b_n + \mathbf{g}_n \cdot (\mathbf{r}_n + \delta\mathbf{r}) \quad (2)$$

$$\varphi_{0,n}(\mathbf{r}) = \varphi_{0,n} + \mathbf{G}_n \cdot \mathbf{r} = \varphi_{0,n} + \mathbf{G}_n \cdot (\mathbf{r}_n + \delta\mathbf{r}) \quad (3)$$

Assuming each imaging voxel has volume $V = a_x a_y a_z$, then $\delta\mathbf{r}_{x,y,z} \in [-0.5a_{x,y,z}, 0.5a_{x,y,z}]$. Thus, by substituting $b(\mathbf{r})$ and $\varphi_0(\mathbf{r})$ using Eqs. (2) and (3) and assuming that the “ideal” signal $\rho(\mathbf{r}; TE)$ only slightly varies across the voxels, the MR signal in Eq. (1) can be rewritten for discrete imaging voxels as follows:

$$\tilde{S}(\mathbf{k}; TE) = \sum_n \sigma_n(TE) \cdot \exp(i\gamma b_n \cdot TE + i\varphi_{0,n}) \cdot \exp(-2\pi i \mathbf{k} \mathbf{r}_n) \cdot P(\mathbf{q}; \mathbf{q}_n; TE) \quad (4)$$

where $\sigma_n(TE)$ is the “ideal” signal from the n^{th} voxel that is free from B_0 -field inhomogeneity artifacts. The function $P(\mathbf{q}; \mathbf{q}_n; TE)$ has the form:

$$P(\mathbf{q}; \mathbf{q}_n; TE) = \prod_{j=x,y,z} \text{sinc}[q_j - q_{n,j}(TE)] \quad (5)$$

For each dimension ($j = x, y, z$), $q_j = k_j a_j = \{-1/2, -1/2 + 1/N_q, \dots, 1/2 - 1/N_q\}$ is the 1D normalized k -space. N_q represents number of q -points used in the calculation. The parameters $q_{n,j}(TE)$ describe phase dispersion across the n^{th} voxel:

$$q_{n,j}(TE) = \frac{1}{2\pi} (g_{\omega,j} \cdot TE + g_{p,j}) \quad (6)$$

where $g_{\omega,j} = \gamma \cdot \mathbf{g}_{n,j} \cdot \mathbf{a}_j$ and $g_{p,j} = \gamma \cdot \mathbf{G}_{n,j} \cdot \mathbf{a}_j$ are the gradients for the n^{th} voxel calculated from the spatially unwrapped frequency ω and phase φ_0 , respectively.

By applying Fourier transform to Eq. (4), the image of mGRE-MRI $S_n(TE)$ can be expressed as follows:

$$S_n(TE) = \frac{1}{N} \cdot \sum_{\mathbf{k}} \tilde{S}(\mathbf{k}; TE) \cdot \exp(2\pi i \mathbf{k} \mathbf{r}_n) = \sum_m \Psi_{nm}(TE) \cdot \sigma_m(TE) \quad (7)$$

where $\Psi_{nm}(TE)$ is a voxel-dependent convolution kernel, which in the 3D case can be presented as follows:

$$\Psi_{nm}(TE) = \exp(i\omega_m \cdot TE + i\varphi_0) \cdot \eta_x \cdot \eta_y \cdot \eta_z \quad (8)$$

where the voxel indices n and m are considered as 3D vectors: (n_x, n_y, n_z) and (m_x, m_y, m_z) , ω_m is the frequency shifts. The factors η_j ($j = x, y, z$) have the form:

$$\eta_j = \sum_q \text{sinc}[q - q_{m,j}(TE)] \cdot \exp[2\pi i q(n_j - m_j)] \quad (9)$$

In some circumstance, Hanning filter is often employed to reduce Gibbs ringing artifacts and improve the signal-to-noise ratio. For images obtained with the Hanning filter, Eq. (9) should be modified by adding an additional factor $\cos^2(\pi q)$. In the 2D case, the factor $\eta_z = \text{sinc}[q_{m,z}(TE)]$.

Using the similarity approximation:

$$\sigma_m(TE) \Rightarrow \sigma_n(TE) \frac{|S_m(0)|}{|S_n(0)|} \quad (10)$$

Eq. (7) reduces to.

$$S_n(TE) = \sigma_n(TE) \cdot F_n(TE) \quad (11)$$

where the F -function, describing the influence of B_0 field inhomogeneity on images, is given by,

$$F_n(TE) = \frac{1}{|S_n(0)|} \sum_m \Psi_{nm}(TE) \cdot |S_m(0)| \quad (12)$$

In practice, the sum in Eq. (12) is taken over several voxels adjacent to a given voxel n . Even for few neighbors included in this sum (as shown previously¹), a large number of voxel-dependent convolutions leads to long computational time. For a typical head scan using mGRE sequence with a matrix size of $256 \times 256 \times 72 \times 10$ (readout, phase encoding, slices, echoes), the calculation time of the F -function in Matlab using 4 neighbors on each dimension could take several hours on a computer with typical configurations.

2.B. Library-driven approach

In the voxel-wise direct calculation, factors η are calculated independently for each voxel. As mentioned above, this procedure usually costs long computational time. In this study, we propose a new way to substantially accelerate the calculation by using a pre-calculated library for η . As shown in Eq. (9), the factors η depend on three parameters: the phase dispersion q_m , neighbor index $(n-m)$, and N_q . When phased-array coils and channel combination methods^{8,9} are used, initial phase φ_0 are eliminated, and the parameter q_m depends only on the frequency gradient g_ω . Relatively large maximum g_ω value should be chosen to sufficiently cover the gradient variations of frequency ω for library building. In our case of imaging the human brain, we found maximum g_ω value of 1500 rads/voxel is sufficient. When using an accuracy of 0.1 rads/voxel, a total number of 30,000 points is sufficient to

build the library for η . Although η also depends on the number of q -points in the sum in Eq. (9), our simulation shows that as long as using a sufficient large $N_q \geq 256$, the factor η is fairly stable (see Fig. 1).

According to Eq. (9), changing the sign of either q_m or neighbor index $\Delta N = (n - m)$ will result in a complex conjugate of η , one can only use non-negative values of q and non-negative neighbor index ΔN for calculations. This can further reduce the computational time for library building. This library of η is then used for F -function calculation: for each voxel and its neighbors with particular g_m values, corresponding values of η can be readily found from the library. Importantly, replacing the direct computation by this new library-driven approach substantially reduces the computing time from hours to a few minutes. As long as including sufficiently large N_q and number of neighboring voxels, the library only needs to be built once and can be used for future calculations. The detailed procedure of library building and F -function calculation is shown in Fig. 2: First, parameters are determined for library building. When using a maximum value of 1500 rads/voxel and an accuracy of 0.1 rads/voxel, a total number of 30,000 points of g_ω ranging from -1500 to 1499.9 rads/voxel are used to build the library for η . For Hanning filtered data, 4 neighbors ($\Delta N = n - m = [-2, -1, 0, 1, 2]$) is enough for each dimension. For non-Hanning filtered data, more neighbors (usually more than 8) are needed. As shown in Fig. 1, $N_q = 256$ can be used for η calculation (step

A in Fig. 2). Second, a library of η is built using the parameters determined in step A, and the indices n_1 and n_2 represent different g_ω values and neighbors ΔN (step B in Fig. 2). At the same time, magnitude and phase images are obtained from the complex dataset (step C in Fig. 2). Magnitude images of the 1st echo are used as an approximation of $|S(0)|$ in Eq. (12). Frequency maps are obtained by performing a linear fitting of the phase images (step D in Fig. 2). Gradients g_ω are calculated from the frequency maps (step E in Fig. 2). Indices n_1 and n_2 are calculated for g_ω and neighbor index ΔN using parameters in step A:

$$n_1 = \frac{g_\omega + 1500}{0.1} + \frac{1}{2} + 1 \quad (13)$$

$$n_2 = \Delta N + 2 + \frac{1}{2} + 1 \quad (14)$$

Indices n_1 and n_2 help to quickly locate η values in the library (step F in Fig. 2), which avoids direct calculation of Eq. (9) and save computational time. Last, η , $|S(0)|$, frequency and g_ω values are used to calculate F -function (step G in Fig. 2).

2.C. Image acquisition and postprocessing

Images were collected from two healthy volunteers (both are 26 years old female) using a 3T MRI scanner (GE Discovery 750w) equipped with a 12-channel phased-array head

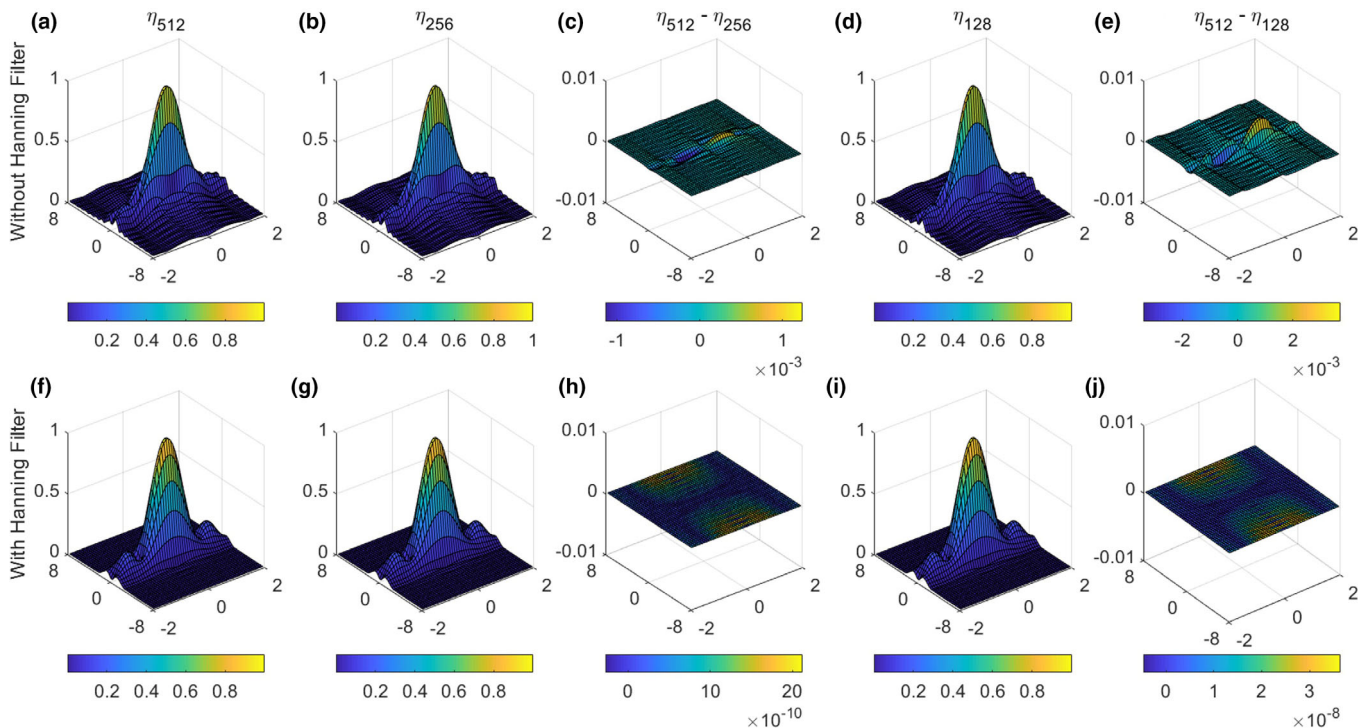


FIG. 1. Simulation of η as a function of the distance from a given voxel to the neighboring voxel (-8 to 8) and the parameter $q_m\epsilon$ ($2, -2$). (a–e): η for the non-filtered data; (f–j): η for the Hanning filtered data. The factor η calculated with $N_q = 512$ (a and f), $N_q = 256$ (b and g) and $N_q = 128$ (d and i) points are shown. The corresponding differences are shown in c, e, h, and j. It turns out that when using a sufficient large number of q -points in the sum in Eq. (9) ($N_q \geq 256$), the changes of the values of η are small ($<1.5 \times 10^{-3}$ for the non-filtered data and $<2.5 \times 10^{-10}$ for the Hanning filtered data). [Color figure can be viewed at [wileyonlinelibrary.com](http://www.wileyonlinelibrary.com)]

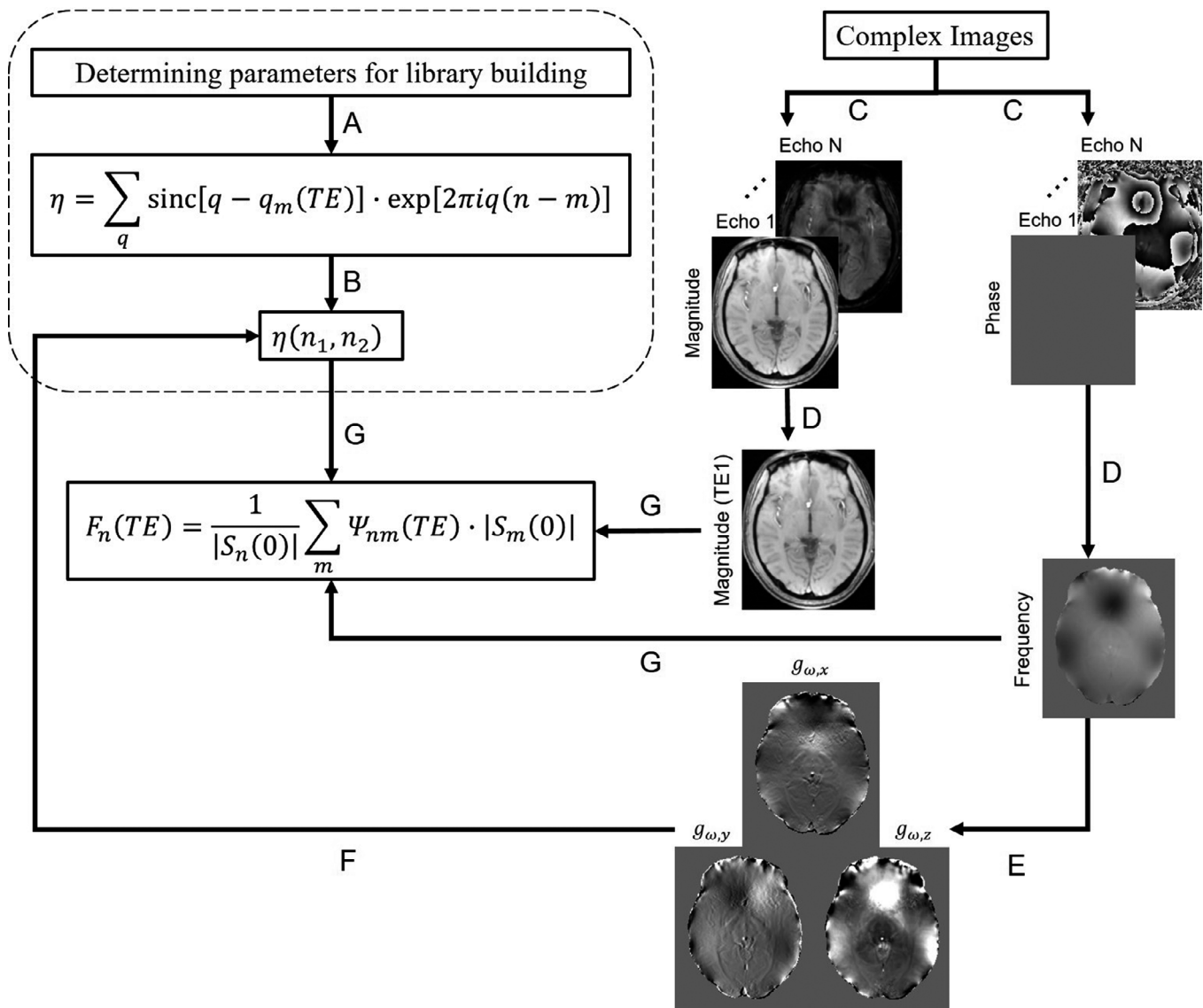


FIG. 2. Flowchart of library building, η values matching and F-function calculation. Library building of η (step A and B) is marked by the dashed box. Magnitudes and phases are obtained from the complex images (step C), and used for later steps (D and G) of F-function calculation. Frequency maps are calculated from the phase images (step D) and used to calculate gradients g_ω (step E). In step F, η values are matched in the library, and used in F-function calculation (step G).

coil. All studies were approved by the IRB of The First Affiliated Hospital of USTC. All the participating healthy volunteers provided written consents.

High-resolution datasets with a voxel size of $1 \times 1 \times 2 \text{ mm}^3$ were acquired using a three-dimensional (3D) mGRE sequence (ESWAN on GE scanners) with a flip angle of 20° and $TR = 50.4 \text{ ms}$. For each acquisition, 10 echoes were collected with first echo time $TE_1 = 3.4 \text{ ms}$ and echo spacing $\Delta TE = 3.8 \text{ ms}$. The total acquisition time was around 11 min. The complex images from different channels were combined using a previously introduced method.^{8,9} The combined mGRE datasets were then used for calculation of the F-function and transverse relaxation rate constant (R_2^*). The libraries were built for three different values of the parameter N_q (512, 256, 128). The F-function was calculated with 4 neighbors on each dimension for Hanning-filtered data and 16 neighbors for non-filtered data.

A maximum value of 1500 rads/voxel and a step of 0.1 rads/voxel were used for the gradient g_m . R_2^* maps were calculated by performing a least square fitting of the mGRE datasets using a mono-exponential model. Previous numerical simulations¹ showed that the field inhomogeneity can distort linear behavior of the signal phase. Hence, we used $q_m < 0.3$ as a criteria to only use short TE for R_2^* fitting in those regions where the field inhomogeneity is severe. All computations were done in Matlab2019a (The MathWorks, Inc., Natick MA, USA) on a computer with standard configurations: 3.2 GHz CPU and 16 GB RAM.

3. RESULTS

The comparison of the F-functions calculated using direct voxel-wise calculation and library-driven approach is shown in Fig. 3. The direct calculation of the F-function takes about

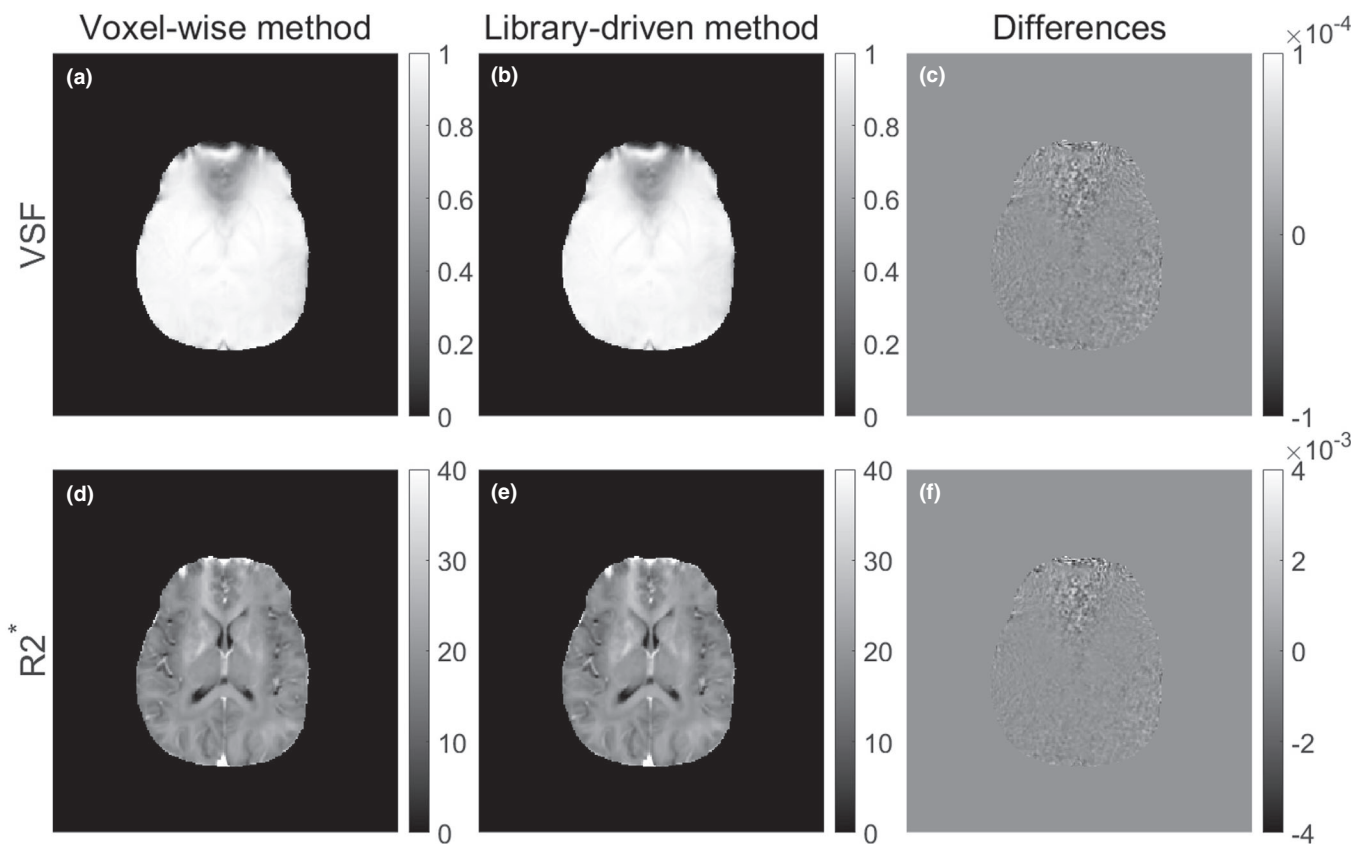


FIG. 3. Comparison of the F -function at TE = 37.5 ms (a–c) and $R2^*$ maps (d–f) using voxel-wise method (a and d) and new library-driven method (b and e). The differences are shown in c and f. The Hanning-filtered images were used to calculate F -function and $R2^*$ maps.

5 hrs and 30 min on a computer with a 3.2 GHz CPU and 16 GB RAM, while the new method requires only 3 min on the same computer. The library building process using parameters introduced in the “Methods” costs less than 30 s on the same computer. The differences between these two methods are below 10^{-4} for the F -function and 10^{-3} for $R2^*$, respectively (panels C, F).

As shown in Fig. 4, we compared $R2^*$ distributions within different ROIs. In the pre-frontal ROI marked by the yellow dashed box, standard deviation (SD) values of $R2^*$

distributions calculated using voxel-wise method and new library-driven method are 4.184 and 4.188 1/s, respectively. In the brain outside yellow dashed box, corresponding SD values of $R2^*$ distributions for the two methods are 4.068 and 4.067 1/s, respectively.

$R2^*$ maps calculated using the F -function with different number of q -points N_q are shown in Fig. 5. Accurate $R2^*$ calculations are guaranteed when sufficient large number of q -points in Eq. (9) ($N_q \geq 256$) are employed, which coincides with the simulation results.

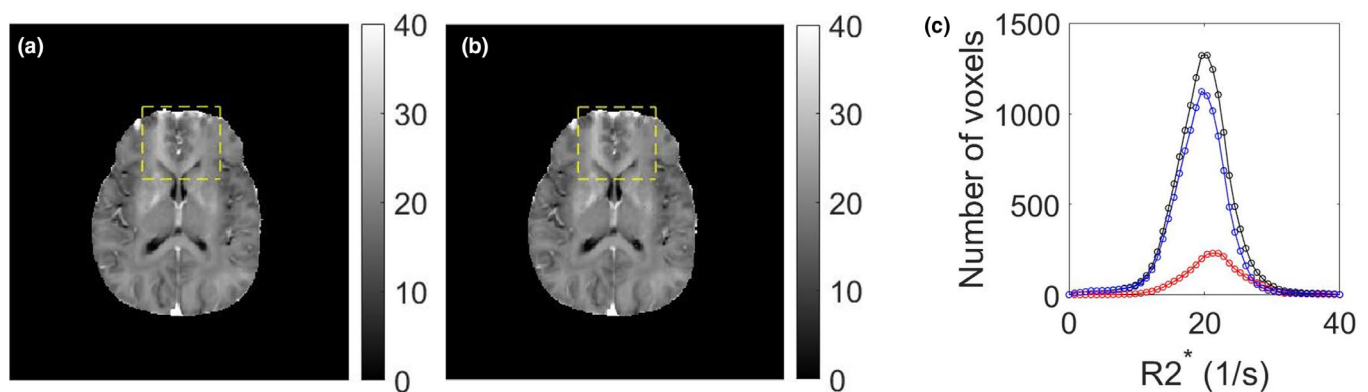


FIG. 4. Comparison of $R2^*$ maps calculated using voxel-wise method (a) and new library-driven method (b). $R2^*$ distributions within different ROIs (Black: brain; Red: brain within yellow dashed box; Blue: brain outside yellow dashed box) are shown in c, with solid lines and circles representing $R2^*$ distribution calculated from a and b, respectively. [Color figure can be viewed at wileyonlinelibrary.com]

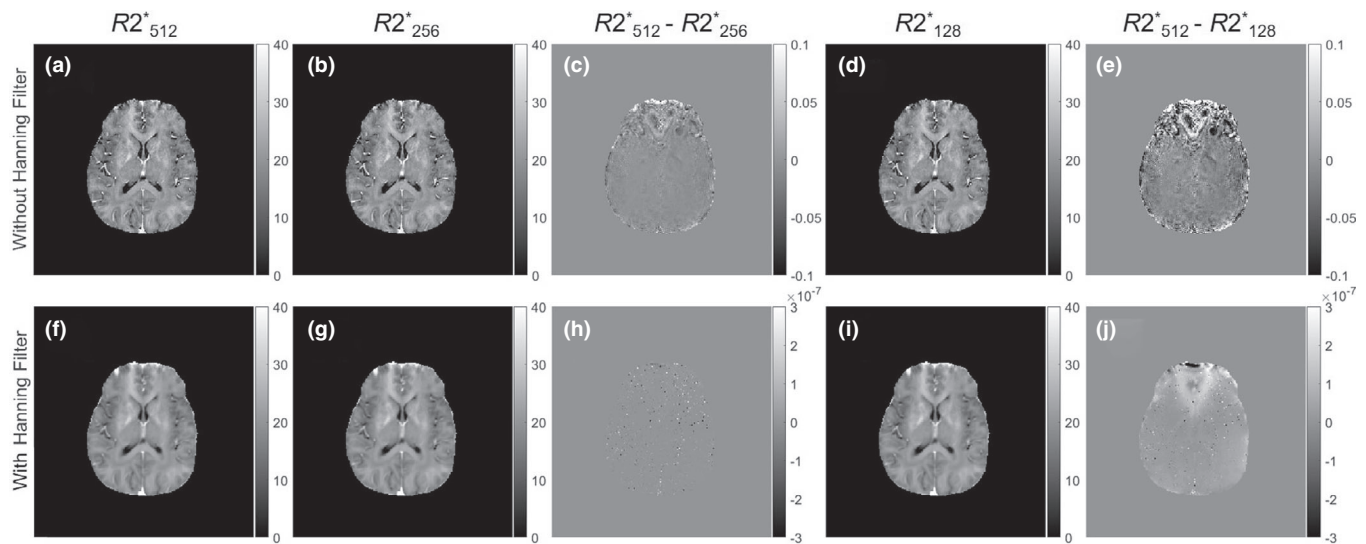


FIG. 5. Comparison of $R2^*$ maps calculated using F-function with different number of q -points N_q (a and f: 512 points; b and g: 256 points; d and i: 128 points). $R2^*$ maps were calculated from non-filtered (a–e) and Hanning-filtered (f–j) data.

4. DISCUSSION AND CONCLUSIONS

Macroscopic B_0 magnetic field inhomogeneity adversely affects MRI images, especially the ones based on a gradient echo acquisition. They not only lead to image distortions,¹⁰ but also degrade quantitative measurements of relaxation parameters^{8,11–15} if not accounted for. Numerous methods have been proposed to correct for these adverse effects, including correcting methods during data collection,^{16–24} and post processing approaches.^{25–30} In this paper we focus on the VSF method¹ that allows accounting for the effects of magnetic field inhomogeneity in data obtained with mGRE sequences.

Quantitative parameters that are derived using mGRE-based techniques, such as $R2^*$ mapping, are usually suffered from magnetic field inhomogeneity artifacts. These artifacts are extremely severe in those regions that are close to the tissue-air interfaces, i.e. in sinuses. These artifacts often deteriorate the quantitative measurements, thus hinder the clinical application of these quantitative techniques. The VSF method was proposed previously to correct for this effect, however, direct voxel-by-voxel calculations of the F -function requires long computational time. In this study, we presented a library-driven approach for fast VSF implementation. By using a pre-calculated library, the calculation time of the F -function has been reduced from 5 hrs and 30 min to 3 min on a computer with a standard configuration (3.2 GHz CPU and 16 GB RAM). By comparing directly with the voxel-wise method, we demonstrated that the library-driven method provided similar accuracy compared with the direct calculation (Fig. 3), even in the high susceptibility area such as pre-frontal regions (Fig. 4).

Based on simulations (Fig. 1) and experiments (Fig. 5), we demonstrated that by selecting a sufficient number of q -points (N_q) and the number of neighboring voxels included in the sum in Eq. (9) to build the library, the new library-driven

method substantially reduces computational time while, at the same time, provides similar accuracy compared with the direct voxel-wise calculation. The reduction of computational time makes it feasible to implement the VSF method with more neighbors, which is necessary for non-filtered data.

We demonstrated the capability of the proposed method to substantially reduce B_0 inhomogeneity artifacts on quantitative $R2^*$ mapping. However, it should be noted that the proposed method is based on certain “linear” approximation [Eqs. (2) and (3)] that is appropriate for high resolution MRI. It might fail to correct low resolution data and data near the edge of an object where the gradients g_m are strong. If the “linear” approximation is not sufficient for calculating effect of field inhomogeneity across the voxel, then Eq. (5) should be substituted by a more general integral that is taken across a single voxel. In this case, the distribution of magnetic field and phases can be found from an independent experiment with higher resolution.

In conclusion, we presented a library-driven approach for fast VSF method implementation. The new procedure provides a fast way to correct mGRE-based images for background field inhomogeneity artifacts, thus can facilitate the applications of mGRE-based quantitative techniques in clinical practices.

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CONFLICT OF INTEREST

The authors have no conflicts to disclose.

5. AVAILABILITY OF DATA

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Ying Liu and Qiong Ye equally contributed to this work.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1. Fast library-driven approach for the two-step iterative calculation of the voxel spread function.